



ANSHUL JAIN

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ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
2026	M.Tech in Optoelectronics & Optical Communication	Indian Institute of Technology Delhi	7.47
2019	B.Sc.(Hons.)	ALIGARH MUSLIM UNIVERSITY	7.61
2014	RBSE	SHREE SK DEORA SCHOOL	79.40
2012	RBSE	SAINT THOM ACADEMY	81.83
2021	M.Sc.	ALIGARH MUSLIM UNIVERSITY	7.88

SCHOLASTIC ACHIEVEMENTS

- Joint CSIR-UGC(NET/JRF) June-Dec 2023, AIR:217
- Graduate Aptitude Test in Engineering (GATE)2024, AIR:431

IIT DELHI THESIS

Title: Beamforming in optical wireless communications

Supervisor: Prof. Swades Kumar De, Communication Networks Research Group Lab., IIT Delhi

Description:- Signal Processing, Adaptive Optics

- A real time Wavefront Correction using Neural Networks and Kalman Filter.
- Study the use of **SLMs**, **DMDs** and **Shack–Hartmann wavefront sensors**

PROJECTS

- **M.Tech. Minor Project- Wireless Optical Communication link Using IR LED and Microcontroller (Prof. Abhishek Dixit, Prof. Amol Choudhary)** (April, 2025) :
 - Developed a prototype for transmitting message signals via IR LED through a 10 m distance with ASCII encoded data.
 - Using **Arduino ide** software on **UART** Protocol & decoding with TSOP1838 receiver (4.5-5.5V, 38kHz).
- **High Capacity FSO upto 300 Gbps using WDM and MDM in the Mid IR Region (Prof. Amol Choudhary)** (Jan, 2025 - May, 2025) :
 - Analysed **WDM & MDM** with **OAM** via **SPP** &- Employed PPLN via DFG/SFG, **EDFA** and **QPSK** Modulation
 - Employed PPLN via DFG/SFG, **EDFA** Amplification and **QPSK** Modulation
 - **Python** Simulation Covered **DFG** efficiency, **OAM** generation and **BER vs OSNR** graph.
- **Diffraction Grating Based OSA (Prof. Amol Choudhary)** (July, 2024 - November, 2024) :
 - Designed and Built a low cost OSA using CMOS cam as detector achieved **1 nm** of resolution in Visible & NIR region.
 - Measured and Displayed Power Distribution of Optical Signal across different wavelength
- **M.Tech. | Optical Communication Lab Experiment (Prof. Abhishek Dixit, Prof. Amol Choudhary)** (2024 - 2025) :
 - Characterization of **DWDM Multiplexer/De-Multiplexer**, **FBG** and **Circulators**.
 - To design and study a *Wireless Optical Communication Link* and find its 3-dB Bandwidth and Voltage vs. Link Length.
 - Design and build a Digital Optical **Fiber Communication** link using **HFBR** transceivers and measure lower and higher data rates & time delay along with **BER** Analysis.
 - Use of **OTDR** to determine fiber break locations and splicing locations in the fiber by observing losses.
 - Designed and experimentally validated a 10-km **IMDD** optical fiber link using a 1310-nm DFB laser with **PRBS** modulation, analysed **Eye Pattern** and **BER** performance
- **Optical Fiber Link Budget Calculator** (Self)(June, 2025 - August,2025): [GitHub Link](#)
 - Developed a **Python-based** tool to simulate Optical Fiber link performance.
 - Calculating **Power budget**, **Dispersion**, **BER** for **BPSK/QPSK/16 QAM** modulation with **Matplotlib** visualization.

TECHNICAL SKILLS

- **Programming and other Languages:** Python | Matlab | C++
- **Packages and Libraries:** NumPy, Pandas, Matplotlib, Seaborn
- **NPTEL Courses** : Python For Data Science, IIT Madras (Prof. Raghunathan Rengasamy)
 - Artificial Intelligence- Search Method for Problem Solving, IIT Madras (Prof. Deepak Khemani)
- **Coursera:** Crash Course on Python by Google

TEACHING ASSISTANTSHIP

- Assisted Students in JOP **Fiber Optics Lab.** with conducting experiments in Laser, **WDM**,**OTDR**,**Optical Fibres**, **EDFA** and **Splicing Machine**.



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IIT COURSE

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COURSES DONE

Fiber Optics, Fiber Optics And Optical Communications Lab., Photonic Devices, Digital Communication & Inform, Optoelectronic Instrumentation, Optical Electronics, Biomedical Optics & Bio-photo., Optical Communication Systems, Micro And Nano Photonics, Fiber Optics And Optical Communication Lab.ii